Math 196 Lecture Notes: Week 1

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1 Introduction

1.1 Course Format and Logistics

1. My name is Miles Kretschmer, I am a PhD student here.
2. My email is `mjkretschmer@uchicago.edu`.
3. My office is Eckhart 7, in the basement. Office hours will be decided this week. Feel free to email if you want to set up a meeting in the meantime.
4. Information about course policies can be found on the syllabus. Briefly, there will be weekly problem sets, one midterm, and a final. The first problem set will be assigned in the next day or two.

1.2 Motivation

Some questions we will be able to answer using the tools of the course include:

1. Solving linear equations: find equilibrium levels of production in an economic model with multiple commodities.
2. Geometry: What is the combined effect of a rotation through $\pi/4$ about the x axis, and a rotation through $\pi/6$ about the y axis?
3. Linear regression: what is the linear function that best fits a set of data points?
4. Dynamical systems: describe the long term evolution of the population sizes of a predator species and a prey species.

1.3 Plan for This Week

We will begin by discussing systems of linear equations in multiple variables. Our goal here will be to learn a method of finding all solutions of such a system of equations. This will be a bit heavy when you see it for the first time; we will have occasion to review it later on, and you will have a chance to practice it on the homework. After this, we will start discussing vectors and their geometric interpretation. This part will be somewhat more conceptual.

2 Linear Equations

A equation is *linear* if it involves only constant multiples of the variables, rather than powers or more complicated functions. Solving linear equations in a single variable is easy enough:

$$ax = b \ \rightsquigarrow \begin{cases} x = b/a & a \neq 0 \\ \text{no solution} & a = 0,\ b \neq 0 \\ x \text{ can be chosen arbitrarily} & a = b = 0 \end{cases}$$

We want a similar description of the solutions of *systems* of linear equations in *multiple variables*. Today we will learn a general method of solving such systems. The best way to explain this is to work through examples, I will later summarize the steps of the method.

Example 1.

$$\begin{cases} x + 2y + 3z = 1 \\ x + 4y + 5z = 0 \\ x + 5y + 5z = 3 \end{cases}$$

The way we will solve systems of equations like this is to add and subtract the equations from eachother to get an equivalent system of the form

$$\begin{cases} x = \dots \\ y = \dots \\ z = \dots \end{cases}$$

To do this systematically, we deal with the variables one at a time. In our example, we start with the x variable. If we subtract the first equation from the other two, we get rid of the x terms in these equations:

$$\begin{cases} x + 2y + 3z = 1 \\ 0x + 2y + 2z = -1 \\ 0x + 3y + 2z = 2 \end{cases}$$

Now everything is as we want with the x variable, so we move to the y variable. If we divide the second equation by 2, we get

$$\begin{cases} x + 2y + 3z = 1 \\ 0x + y + z = -1/2 \\ 0x + 3y + 2z = 2 \end{cases}$$

Now we can subtract 2 times the second equation from the first, and 3 times it from the third equation, to get

$$\begin{cases} x + 0y + z = 2 \\ 0x + y + z = -1/2 \\ 0x + 0y - z = 7/2 \end{cases}$$

Now we are done with the y variable, so we turn to the z variable. Multiplying the last equation by -1, we get

$$\begin{cases} x + 0y + z = 2 \\ 0x + y + z = -1/2 \\ 0x + 0y + z = -7/2 \end{cases}$$

Then subtracting it from the first second equations, we get

$$\begin{cases} x + 0y + 0z = 11/2 \\ 0x + y + 0z = 3 \\ 0x + 0y + z = -7/2 \end{cases}$$

As desired. This system of equations has a unique solution.

2.1 Augmented Matrix Notation and Row Operations

For a more concise representation of these operations, we can put all of the coefficients of the equations into an array, and carry out our operations directly on this array. For example, the system

$$\begin{cases} 2x + 4y - z = 2 \\ -x - 2y + z = 1 \\ 3x + 7y + 2z = 4 \end{cases}$$

is represented by

$$\left[\begin{array}{ccc|c} 2 & 4 & -1 & 2 \\ -1 & -2 & 1 & 1 \\ 3 & 7 & 2 & 4 \end{array}\right]$$

In general, we call an array of numbers like this a *matrix.* This is called the *augmented matrix* of the linear system, the part before the line is the *coefficient matrix.*

Previously, we simplified systems of equations by multiplying both sides of an equation by constants, or adding a multiple of one equation to another. In this format, we use the corresponding operations on the rows of the matrix. To this end, we can define operations on rows of numbers as follows:

$$\begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} + \begin{bmatrix} b_1 & \dots & b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 & \dots & a_n + b_n \end{bmatrix}$$

$$c \cdot \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} = \begin{bmatrix} ca_1 & \dots & ca_n \end{bmatrix}$$

Later on, we will understand these operations as vector addition and scalar multiplication, but already we can see that they correspond to operations on equations. The specific operations we use to simplify systems of equations consist of the following:

1. Multiplying a row by a nonzero constant.
2. Adding a multiple of one row to another.
3. Changing the order of the rows,

We call these *elementary row operations.*

We now work one column at a time, moving from left to right, with the goal of transforming the matrix to have the form

$$\left[\begin{array}{ccc|c} 1 & & & * \\ & \ddots & & \vdots \\ & & 1 & * \end{array}\right]$$

We will later know the coefficient matrix here as the *identity matrix.* For our example, the sequence of steps is as follows:

$$\left[\begin{array}{ccc|c} 2 & 4 & -1 & 2 \\ -1 & -2 & 1 & 1 \\ 3 & 7 & 2 & 4 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 2 & -1/2 & 1 \\ -1 & -2 & 1 & 1 \\ 3 & 7 & 2 & 4 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 2 & -1/2 & 1 \\ 0 & 0 & 1/2 & 2 \\ 0 & 1 & 7/2 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 2 & -1/2 & 1 \\ 0 & 1 & 7/2 & 1 \\ 0 & 0 & 1/2 & 2 \end{array}\right] \rightsquigarrow$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -15/2 & -1 \\ 0 & 1 & 7/2 & 1 \\ 0 & 0 & 1/2 & 2 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & -15/2 & -1 \\ 0 & 1 & 7/2 & 1 \\ 0 & 0 & 1/2 & 2 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & -15/2 & -1 \\ 0 & 1 & 7/2 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & -13 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

Translating back into equations, we get $x = 29$, $y = -13$, $z = 4$.

2.2 Systems that do not have a unique solution

It will not always be possible to transform the matrix into one of the form

$$\left[\begin{array}{ccc|c} 1 & & & * \\ & \ddots & & \vdots \\ & & 1 & * \end{array}\right]$$

This is because some systems do not have a solution, or have more than one solution.

Example 2. Consider the system of equations

$$\begin{cases} 2x + 4y = 2 \\ -x - 2y = 1 \end{cases}$$

corresponding to the augmented matrix

$$\left[\begin{array}{cc|c} 2 & 4 & 2 \\ -1 & -2 & 1 \end{array}\right]$$

Applying our method, we divide the first row by 2, and add it to the second:

$$\left[\begin{array}{cc|c} 2 & 4 & 2 \\ -1 & -2 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ -1 & -2 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right]$$

We cannot proceed from here, because we do not have nonzero terms in the second row where we need them. Translating this back to equations, the second row is $0 = 2$, which is inconsistent. The conclusion we draw from this is that the original system has no solutions. [Draw lines picture]

What if, on the other hand, our second equation was $-x - 2y = -1$? Then applying the same steps, we get

$$\left[\begin{array}{cc|c} 2 & 4 & 2 \\ -1 & -2 & -1 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ -1 & -2 & -1 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 0 & 0 \end{array}\right]$$

Still, there is nothing more we can do. Now our second equation is $0 = 0$, which is always true, and our first is $x + 2y = 1$. The conclusion we draw from this is that we can choose the value of y arbitrarily, and then x is determined by $x = 1 - 2y$. Thus our system of equations has an infinite set of solutions. [The lines coincide]

Example 3. Here is another way our process can fail to give us a unique solution. Consider the system

$$\begin{cases} x + y + z = 0 \\ x + y - 2z = 4 \end{cases}$$

which has more variables than equations. This corresponds to the matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & -2 & 4 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & -3 & 4 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & -4/3 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 4/3 \\ 0 & 0 & 1 & -4/3 \end{array}\right]$$

This translates to $z = -4/3$, $x + y = 4/3$. Again, y can be chosen arbitrarily, and we get $x = 4/3 - y$, yielding infinitely many solutions.

2.3 Reduced Row Echelon Form

In these cases, the final form of the matrix still gave us a description of all the solutions of the system, even if there was no solution or if there were infinitely many solutions.

Definition. A coefficient matrix is said to be in *reduced row echelon form* if

1. Each row is either all 0s or has 1 as its first nonzero entry.
2. As we move down row by row, the leading 1 moves to the right, and all 0 rows are at the bottom.
3. Each column containing a leading 1 has all its other entries 0

$$\begin{bmatrix} 0 & 1 & * & 0 & * \\ 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

We call the columns with a leading 1 the *pivot columns*.

If the coefficient matrix is in row reduced echelon form, then we can find all of solutions of the system using *backwards substitution*: we start with the last variable and work backwards. At each step, if the variable corresponds to a non-pivot column, it can be chosen arbitrarily. If the variable corresponds to a pivot column, the row containing the 1 lets us solve for this variable in terms of our previous choices.

Example 4.

$$\left[\begin{array}{ccccc|c} 0 & 1 & 2 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 3 \end{array}\right]$$

Has coefficient matrix in reduced row echelon form. This corresponds to equations

$$\begin{cases} x_2 + 2x_3 - x_5 = 2 \\ x_4 + x_5 = 3 \end{cases}$$

Using backwards substitution, we can describe the solutions as follows. The variable x_5 is a non-pivot, so $x_5 = r$ is arbitrary. The variable x_4 corresponds to a pivot column with leading 1 in the second row, and using this we solve for $x_4 = 3 - x_5 = 3 - r$. The variable x_3 is a non-pivot, so $x_3 = s$ is arbitrary. The variable x_2 corresponds to a pivot column: using the first row, $x_2 = 2 - 2x_3 + x_5 = 2 + r - 2s$. Finally, x_1 can be chosen arbitrarily, $x_1 = t$. Therefore, we have infinitely many solutions, which can be explicitly described as

$$\begin{cases} x_1 = t \\ x_2 = 2 + r - 2s \\ x_3 = s \\ x_4 = 3 - r \\ x_5 = r \end{cases}$$

for arbitrary choices of r, s, t.

Example 5.

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 3 \end{array}\right]$$

has coefficient matrix in reduced row echelon form. The final row corresponds to $0 = 3$, so the system has no solutions. On the other hand, in

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The last row is $0 = 0$, which is always true. We set $x_3 = t$ arbitrary, and get $x_2 = -1 - t$, $x_1 = 2t$.

Given any coefficient matrix, we can always use row operations to transform it so that to reduced row echelon form. The algorithm to do this is what we have been doing in all the examples. We work one column at a time. If the column contains the leading nonzero entry of some row, we can make it a pivot column. First, divide through by this nonzero entry to make it 1, then use this 1 to eliminate all other entries in the column by subtracting multiples of this row. Then swap rows to move the row up to the appropriate place, below the other rows with leading 1s. If the column does not contain the leading nonzero entry of a row, it will not be a pivot column, and we move on.

Remark. When we are applying this method to solve a linear equation, we only consider the columns of the coefficient matrix, left of the line. The entries to the right of the line just come along for the ride.

Example 6. Consider the matrix

$$\begin{bmatrix} 3 & 6 & 3 & 0 \\ 2 & 4 & 2 & 4 \\ -1 & -2 & 2 & -3 \end{bmatrix}$$

We put it into reduced row echelon form as follows:

$$\begin{bmatrix} 3 & 6 & 3 & 0 \\ 2 & 4 & 2 & 4 \\ -1 & -2 & 2 & -3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 4 \\ -1 & -2 & 2 & -3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 3 & -3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 1 & -1 \end{bmatrix} \rightsquigarrow$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 1 & -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note that here I am only looking at the coeficient matrix, I have left off the additional column for simplicty.

2.4 Summary of Gaussian Elimination, Row Operations, and Echelon Form

To summarize our method for solving linear equations, we proceed in three steps:

1. Translate the system of equations into an augmented matrix.
2. Apply row operations to put the coefficient matrix in reduced row echelon form, dealing with one column at a time, making the column a pivot whenever possible.
3. Translate back into linear equations, obtain the solutions using backwards substitution.

This method of solving linear systems is called *Gaussian Elimination*. Applying this method, we have seen that linear systems can have a unique solution, infinitely many solutions, or no solutions at all, depending on the shape of the reduced row echelon form.

3 Vectors

We will now temporarily set aside our method of solving linear equations, and introduce a geometric language which allows us to understand what it is we are doing when we solve them. The main objects of this language are *vectors*. We will think of these in two ways. One way of thinking of a vector is as a list of numbers, which we will write as a column:

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The other way of thinking of a vector is as an arrow in space. A vector with 2 entries corresponds to an arrow in the plane, pointing from the origin to the point indicated by its entries. A vector with 3 entries corresponds to an arrow in 3 dimensional space.

We let $\mathbb{R}^n$ denote the collection of all vectors with n entries, so we can think of $\mathbb{R}^2$ as the plane, $\mathbb{R}^3$ as space, and so on.

3.1 Operations on Vectors

We can define algebraic operations on vectors, which we will now try to understand in both a geometric way and in terms of lists of numbers.

3.1.1 Vector Addition

To add two vectors v and w, thought of as arrows in the plane, we can arrange them so that v starts at the origin, and w starts at the tip of v. Then the sum $v + w$ is the arrow pointing from the origin to the tip of this shifted version of w. Notice that we get the same answer if we instead put w at the origin and shift v.

In terms of a list of numbers, this corresponds to the formula

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix}$$

This is the same way we added rows when solving linear equations.

Example 7.

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Both of these procedures make sense in 3 dimensions as well. Adding columns entry by entry makes sense in any number of dimensions, even if we can no longer visualize the arrows.

3.1.2 Scalar Multiplication

The other operation we will apply to vectors is multiplication of a vector by a number, $c \cdot v$. Geometrically, if $c > 0$, $c \cdot v$ points in the same direction as v, but its length is multiplied by c. If $c < 0$, it points in the opposite direction. If $c = 0$, its length is 0. [Draw picture]. If we fix v and let c vary, the vectors $c \cdot v$ sweep out a line through the origin in the direction of v. We call this process *scaling* the vector v, and refer to the number c as a *scalar*. In general scalar and number (as opposed to vector) are synonymous.

Algebraically, we have the formula:

$$c \cdot \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} ca_1 \\ ca_2 \end{bmatrix}$$

Example 8.

$$-2 \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

Again, this process makes sense in 3 dimensions as well, and the formula makes sense in any number of dimensions. In 3 dimensions, scaling a (nonzero) vector with all possible scalars still sweeps out a line through the origin.

3.2 Linear Combinations

Given two vectors v_1 and v_2, a *linear combination* of v_1 and v_2 is an expression

$$c_1 \cdot v_1 + c_2 \cdot v_2$$

For scalars c_1, c_2. Geometrically, we can think of this as the result of traveling in the v_1 direction for a distance set by c_1, and in the v_2 direction for a distance set by c_2.

Example 9. In 3 dimensional space, we can consider all the linear combinations of the vectors

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},\ e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

We have

$$a \cdot e_1 + b \cdot e_2 = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ b \\ 0 \end{bmatrix}$$

By choosing a and b appropriately, any vector whose last entry is 0 can be expressed this way. Geometrically, these vectors sweep out a plane through the origin, defined by the equation $z = 0$ in coordinates x, y, z.

If we add to e_1 and e_2 the vector

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

then we can write any vector in 3 dimensional space as a linear combination of e_1, e_2, e_3:

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = a \cdot e_1 + b \cdot e_2 + c \cdot e_3$$

We call e_1, e_2, e_3 the *standard basis* of $\mathbb{R}^3$. Later on we will learn what a basis is in general.

Question. Can we write

$$\begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

as a linear combination of

$$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix},\ w = \begin{bmatrix} 2 \\ -1 \end{bmatrix}?$$

We are looking for numbers x, y so that

$$\begin{bmatrix} 2 \\ -3 \end{bmatrix} = x \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} x + 2y \\ x - y \end{bmatrix}$$

If we compare the entries of the vectors, we get a system of equations:

$$\begin{cases} x + 2y = 2 \\ x - y = -3 \end{cases}$$

Let's apply our method to solve this. We translate it into an augmented matrix

$$\left[\begin{array}{cc|c} 1 & 2 & 2 \\ 1 & -1 & -3 \end{array}\right]$$

Notice that the columns of the coefficient matrix are v and w, and the column to the right of the line is the goal vector. Applying row operations, get get

$$\left[\begin{array}{cc|c} 1 & 2 & 2 \\ 1 & -1 & -3 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & 2 \\ 0 & -3 & -5 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & 2 \\ 0 & 1 & 5/3 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & -4/3 \\ 0 & 1 & 5/3 \end{array}\right]$$

This translates back to $x = -4/3$, $y = 5/3$, and so we can write

$$\begin{bmatrix} 2 \\ -3 \end{bmatrix} = -4/3 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 5/3 \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

[Draw this]

Question. Can we write

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

as a linear combination of

$$v = \begin{bmatrix} 1 \\ 2 \end{bmatrix},\ w = \begin{bmatrix} -2 \\ -4 \end{bmatrix}?$$

If we set this up as a system of equations, the augmented matrix will be

$$\left[\begin{array}{cc|c} 1 & -2 & 0 \\ 2 & -4 & 1 \end{array}\right]$$

Applying a row operation, we get

$$\left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 1 \end{array}\right]$$

The second row corresponds to the equation $0 = 1$, so this system is inconsistent. The answer to our question is no.

We can explain this geometrically: we have $w = -2 \cdot v$, so a linear combination

$$a \cdot v + b \cdot w = a \cdot v + b \cdot (-2 \cdot v) = (a - 2b) \cdot v$$

Thus any linear combination of v and w can be written as just a scalar multiple of v. These sweep out a line in the plane, and the vector

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

does not lie on this line. [Draw picture]

Just as we can translate the problem of writing one vector as a linear combination of others into a linear system, we can go the other way, and view any linear system as asking for a way of writing one vector as a linear combination of others. In the augmented matrix, these vectors are exactly the columns.

4 Applying a Matrix to a Vector

We have so far considered systems of linear equations, and the problem of writing one vector as a linear combination of others, and seen that they are equivalent. In both of these cases, we can view a solution to the problem as a vector as well. It is, after all, a list of numbers. In the second to last example, our solution was the vector

$$\begin{bmatrix} -4/3 \\ 5/3 \end{bmatrix}$$

In general, viewing the solution to the problem as a vector, we can understand these problems by introducing a new operation on vectors. Recall that a matrix is a rectangular array of numbers:

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$$

We can view its columns as vectors (with m entries), so we write

$$A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

Given a vector with n entries:

$$w = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

we define the matrix vector product Aw to be the linear combination

$$Aw = x_1 \cdot v_1 + \dots + x_n \cdot v_n$$

Example 10.

$$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = 1 \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 3 \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} + \begin{bmatrix} -3 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

In order for a matrix-vector product to be defined, the vector needs to have as many entries as there are columns in the matrix, so

$$\begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

is makes sense, and yields a vector with 3 entries, but

$$\begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

is undefined. The number of columns (width) of the matrix is the size of an "input" vector, and the number of rows (height) is the size of an "output" vector.

Conclusion. *Suppose $v_1, \dots, v_n$, $v_\star$ are vectors with m entries. We let*

$$A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

be the matrix with columns $v_1, \dots, v_n$. The following 3 problems are equivalent:

1. *Solve the system of linear equations with augmented matrix*

$$\left[A \mid v_\star\right]$$

2. *Write $v_\star$ as a linear combination of $v_1, \dots, v_n$.*
3. *Find a vector $w \in \mathbb{R}^n$ such that $Aw = v_\star$.*

Given a matrix A, we can view it as a function which takes vectors as input, and returns a vector as output, via the matrix vector product:

$$v \mapsto Av$$

Next week, we will discuss algebraic and geometric properties of these functions.